Springboard Virtual Internship 6.0

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Domain: Data Visualization

Project: Visualizing U.S. Natural Disaster Declarations

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Milestone-02: Temporal Trend Analysis

1.Objective

The objective of Milestone-02 is to analyze how disaster declarations evolve over time using the FEMA Disaster Declarations dataset. The focus of this milestone is to:

- Identify overall temporal trends
- Compare trends across different incident types
- Detect seasonal patterns
- Interpret meaningful insights from visualizations

This analysis helps in understanding the historical behavior of disaster occurrences in the United States.

2.Dataset Overview

The analysis is based on the FEMA Disaster Declarations Summaries dataset, which contains records of federally declared disasters.

Key Columns Used:

- declarationDate – Date of disaster declaration
- incidentType – Type of disaster (e.g., Flood, Hurricane, Fire)
- state – Affected state

The dataset was cleaned and prepared for time-based analysis.

3.Time Data Validation

Before performing temporal analysis, the date column was validated to ensure accuracy.

Actions Performed:

- Converted declarationDate into datetime format

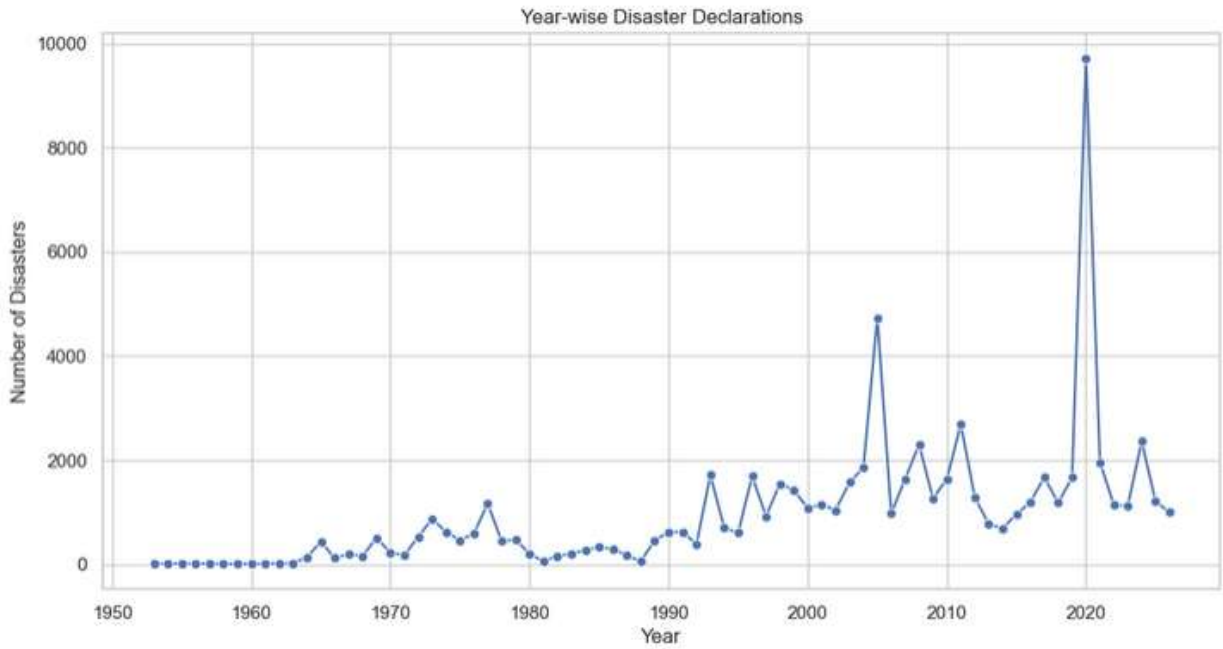
- Extracted:
    - year
    - month
- Checked for missing or invalid date values

Importance:

Correct date formatting is essential, as incorrect time data can lead to invalid trend analysis.

4.Yearly Trend Analysis

The total number of disaster declarations per year was calculated and visualized using a line chart.



Interpretation:

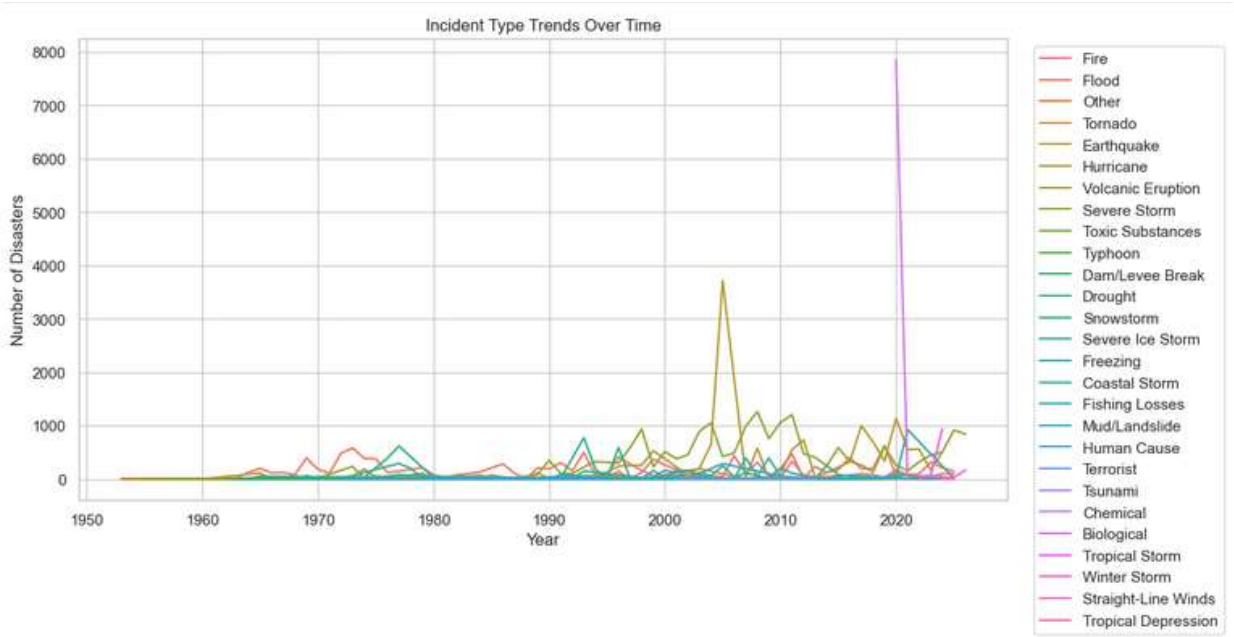
1. The graph shows the total number of disaster declarations across years.
2. An overall increasing trend can be observed in recent decades.
3. Certain years show noticeable spikes, which may correspond to major nationwide disaster events or policy changes in declaration processes.

This suggests that disaster frequency has increased over time, possibly due to climate patterns, environmental changes, or improved reporting systems.

5.Incident Type Trends Over Time

To understand how different disaster types evolve over time, data was grouped by year and incidentType.

A multi-line chart was created to compare trends across major disaster categories.



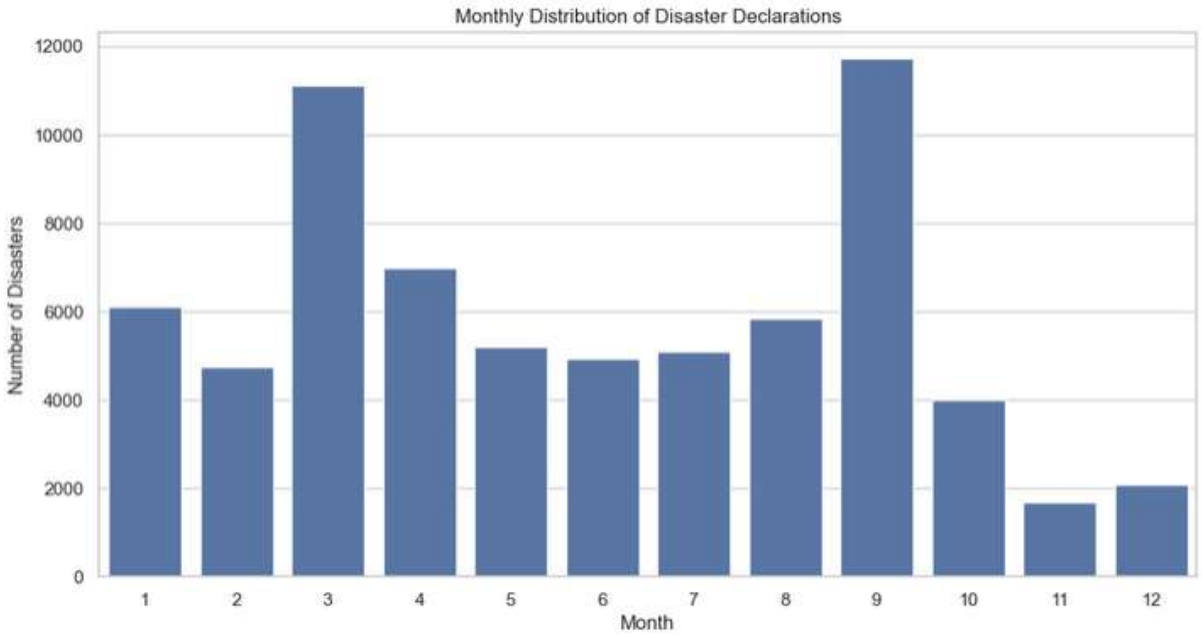
Interpretation:

1. The graph compares various disaster types over multiple years.
2. Certain incident types such as storms and floods appear more frequently than others.
3. Some disaster types show steady growth trends, indicating increasing occurrence over time.
4. A few categories remain relatively stable or fluctuate depending on specific years.

This comparison highlights which disaster types contribute most to the overall increase in declarations.

6.Seasonality Analysis (Monthly Trends)

To detect seasonal patterns, disasters were grouped by month and visualized using a bar chart.



Interpretation:

1. The graph displays the distribution of disaster declarations across months.
2. Certain months show higher disaster activity compared to others.
3. This indicates seasonal concentration, which may align with hurricane seasons, wildfire seasons, or other weather-related cycles.

The seasonal pattern suggests that disasters are not evenly distributed throughout the year.

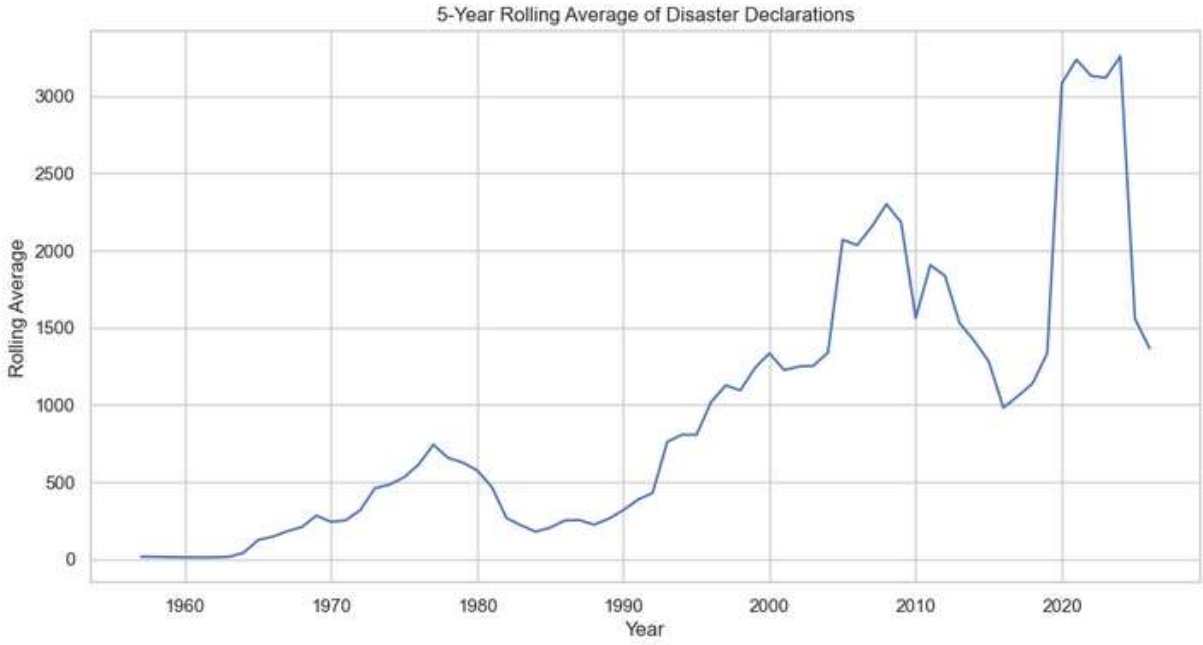
7.Advanced Time Insights (Optional Enhancements)

7.1 Rolling Average Analysis

A 5-year rolling average was applied to smooth short-term fluctuations.

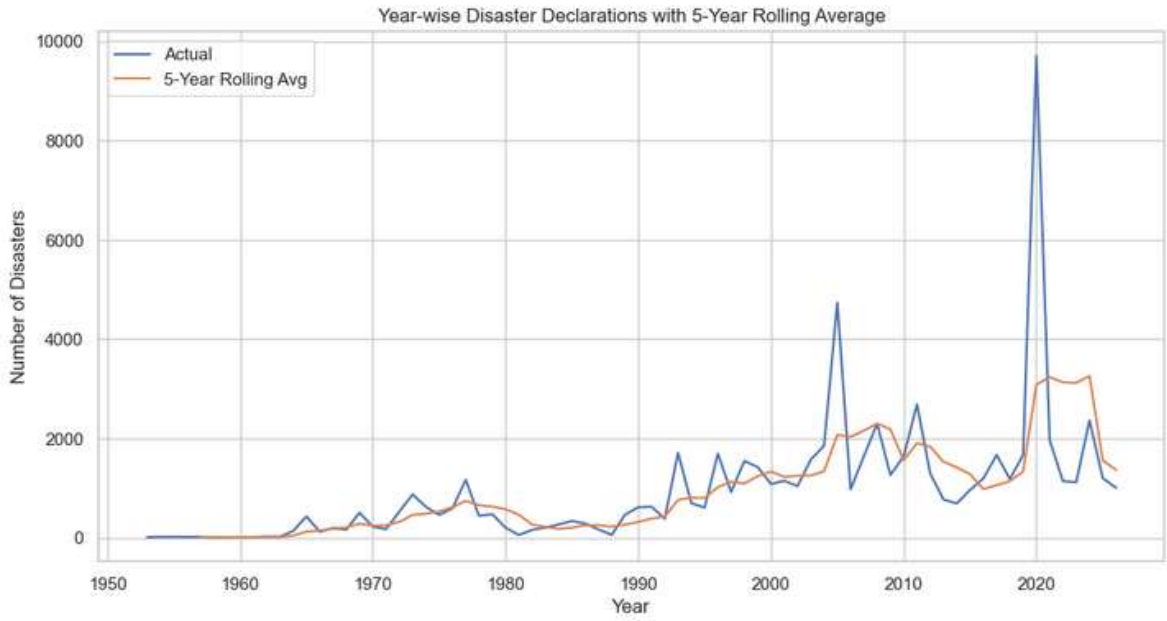
Insight:

- The rolling average highlights the long-term trend more clearly.
- It confirms a gradual upward movement in disaster declarations over time.



7.2 Peak Year Detection

The year with the highest number of disaster declarations was identified.



Insight:

- The peak year represents a period of unusually high disaster activity.
- This may correspond to extreme climatic events or multiple large-scale disasters occurring in the same year.

8.Summary of Findings

The temporal analysis of disaster declarations reveals:

- A noticeable increase in disaster declarations over time
- Certain disaster types dominate the dataset
- Seasonal patterns exist within specific months
- Long-term trend smoothing confirms upward movement

These insights provide a strong foundation for further geographical and relational analysis in upcoming milestones.

9.Conclusion

Milestone-02 successfully identifies temporal trends and seasonal behavior in FEMA disaster declarations. The analysis demonstrates how disaster frequency and type evolve over time, offering meaningful insights into historical patterns.

This structured temporal analysis serves as a critical step toward understanding disaster dynamics in the United States.